UFSM00066 · Sistemas Dinâmicos
Lista de exercícios 2
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a) $y'(t) + 4y(t) = r(t);$

1. $sY(s) + 4Y(s) = R(s)$

$Y(s)(s+4) = R(s)$

$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{s+4}$

$\delta(t)$

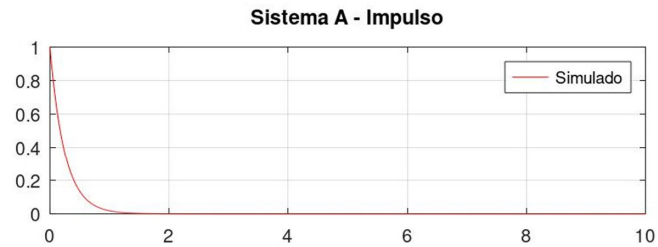
$R(s) = 1$

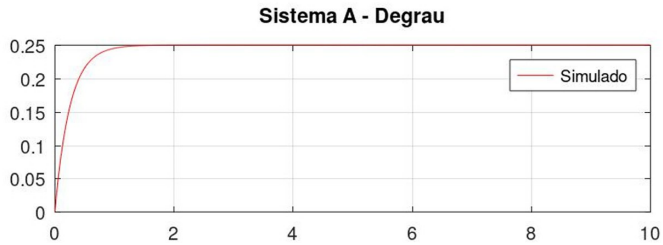
$Y(s) = \frac{1}{s+4}$

$u(t)$

$R(s) = \frac{1}{s}$

$Y(s) = \frac{1}{s(s-4)}$





2. $y(t) = e^{-4t}$

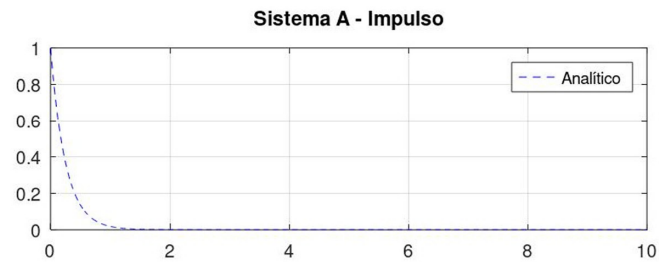
$\frac{1}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$

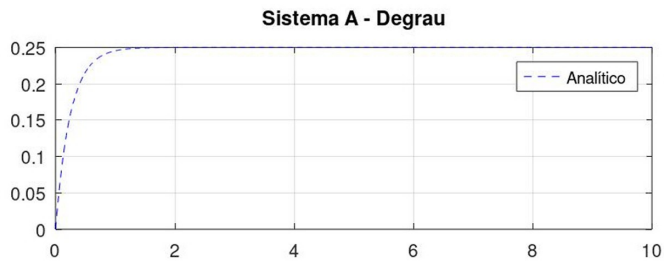
$1 = A(s+4) + Bs$

$A = 1/4 \quad B = -1/4$

$Y(s) = \frac{1}{4s} - \frac{1}{4(s+4)}$

$y(t) = \frac{1}{4}(1 - e^{-4t})$





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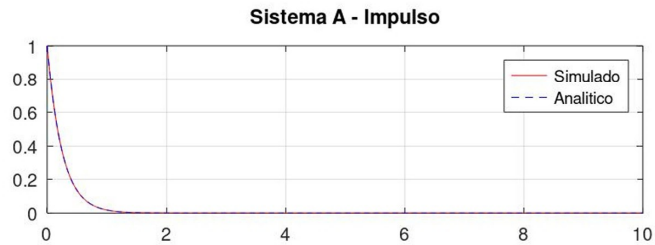


b) $y''(t) + 5y'(t) + 4y(t) = r(t);$

1. $s^2 Y(s) + 5s Y(s) + 4Y(s) = R(s)$

$Y(s)(s^2+5s+4) = R(s)$

$(s+1)(s+4)$

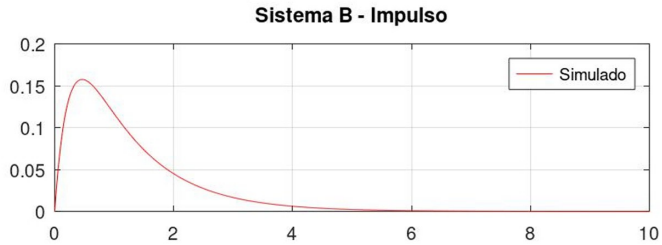
$\delta(t)$

$Y(s) = \frac{1}{(s+1)(s+4)}$

$\frac{1}{3}\left(\frac{1}{s+1} - \frac{1}{s+4}\right)$

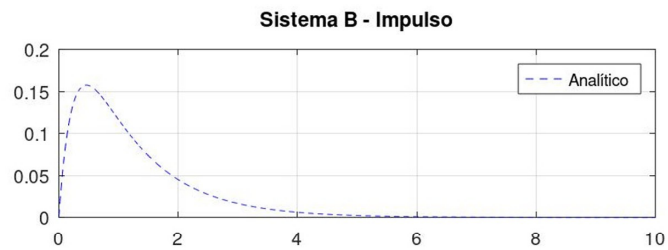
$u(t)$

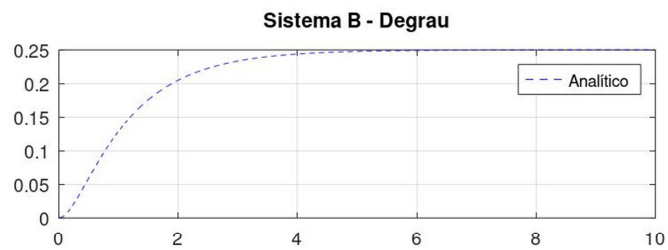
$Y(s) = \frac{1}{s(s+1)(s+4)}$



2. $y(t) = \frac{1}{3}(e^{-3} - e^{-4t})$

$y(t) = \frac{1}{4} - \frac{1}{3}e^{-t} + \frac{1}{12}e^{-4t}$





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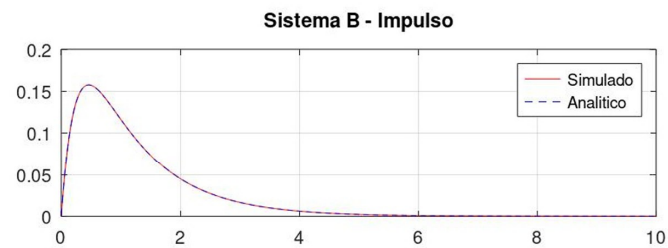
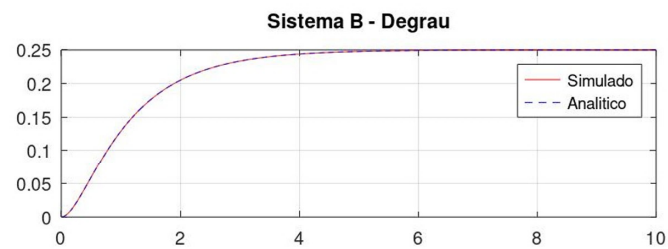




c) $y''(t) - 4y'(t) + 4y(t) = r(t);$

1. $G(s) = \dfrac{1}{s^2 - 4s + 4}$

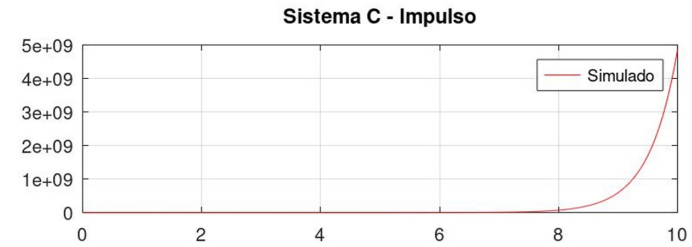
$(s-2)^2$

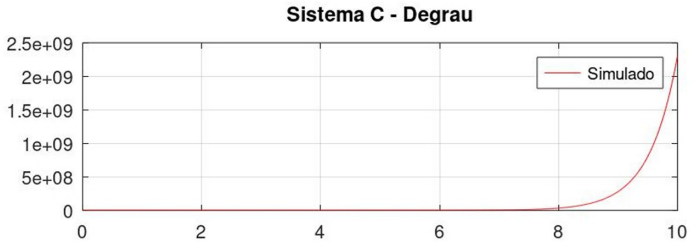
$\delta(t)$

$Y(s) = \dfrac{1}{(s-2)^2}$

$u(t)$

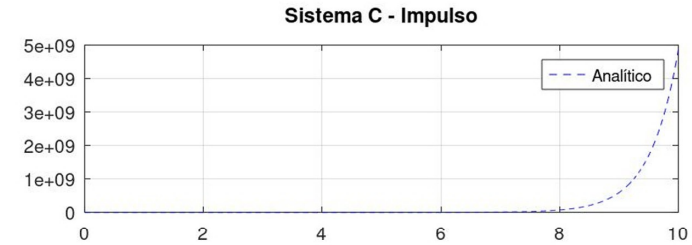
$Y(s) = \dfrac{1}{s(s-2)^2}$

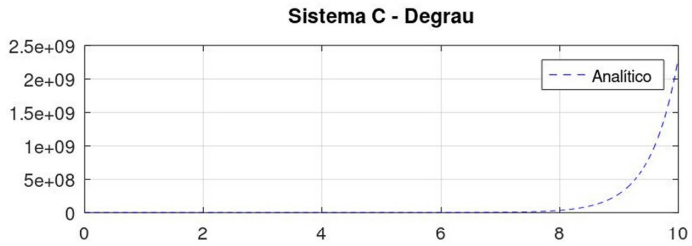




2. $y(t) = te^{2t}$

$y(t) = \dfrac{1}{4} - \dfrac{e^{2t}}{4} + \dfrac{te^{2t}}{2}$





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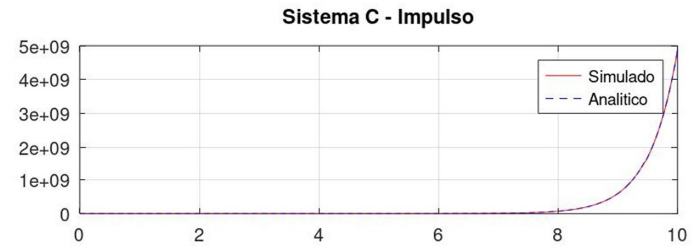
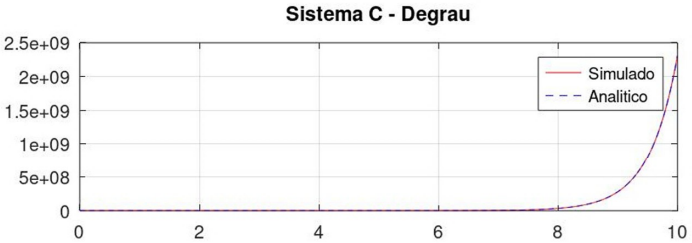




d) $y''(t) - y'(t) = r(t)$;

1. $s^2 Y(s) - sY(s) = R(s)$

$Y(s)(s^2 - s) = R(s)$

$G(s) = \frac{1}{s(s-1)}$

$\delta(t)$

$Y(s) = \frac{1}{s(s-1)}$

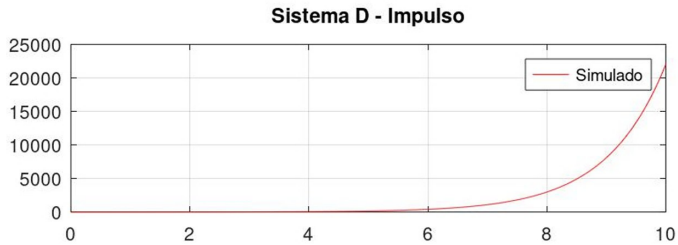
$Y(s) = \frac{-1}{s} + \frac{1}{s-1}$

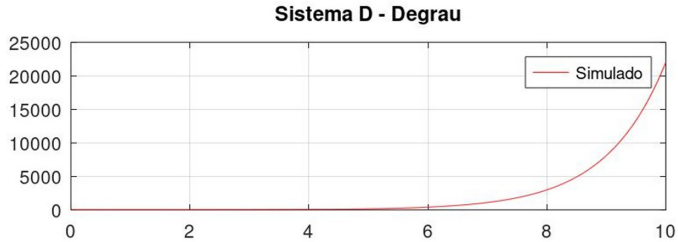
$u(t)$

$Y(s) = \frac{1}{s} \cdot \frac{1}{s(s-1)}$

$\frac{1}{s^2(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1}$

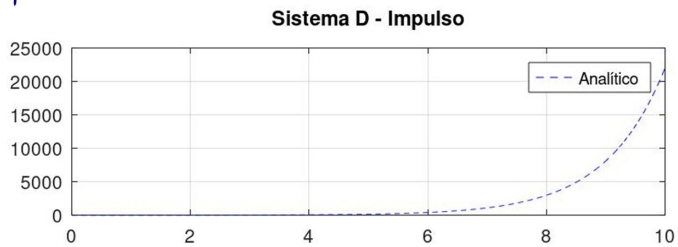
$Y(s) = \frac{-1}{s} - \frac{1}{s^2} + \frac{1}{s-1}$

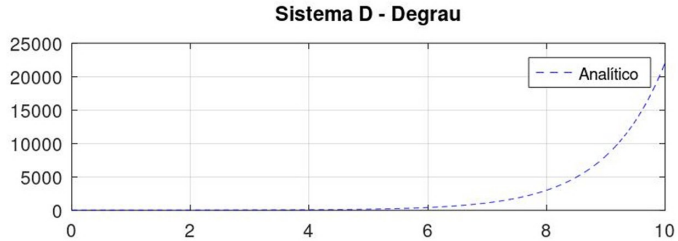




2. $y(t) = e^t - 1$

$y(t) = e^t - 1 - t$





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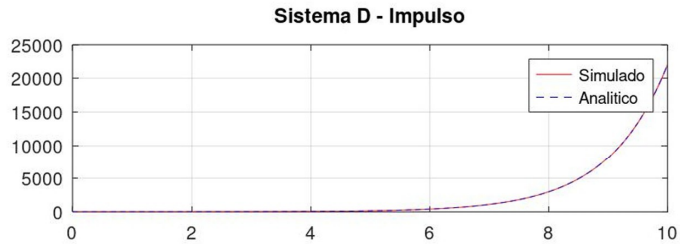
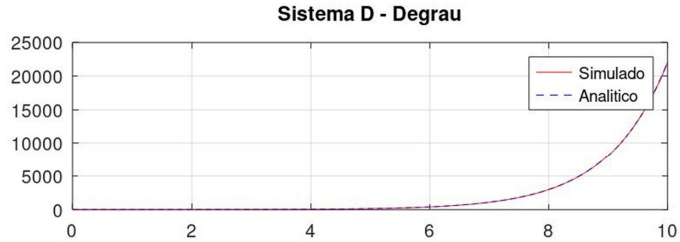




e) $y''(t) + 4y'(t) = r(t);$

1. $s^2 Y(s) + 4sY(s) = R(s)$

$Y(s)(s^2+4s) = R(s)$

$$G(s) = \frac{1}{s^2+4s} = \frac{1}{s(s+4)}$$

$\delta(t)$

$$Y(s) = \frac{1}{s(s+4)}$$

$$\frac{1}{s(s+4)} = \frac{A}{s} + \frac{B}{s+4}$$

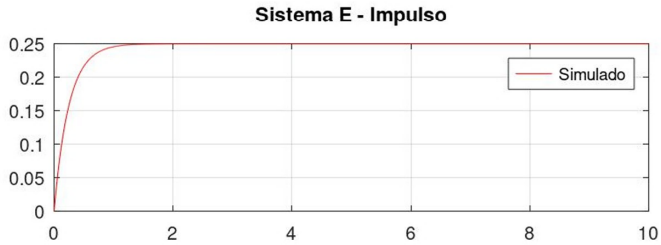
$$Y(s) = \frac{1}{4s} - \frac{1}{4(s+4)}$$

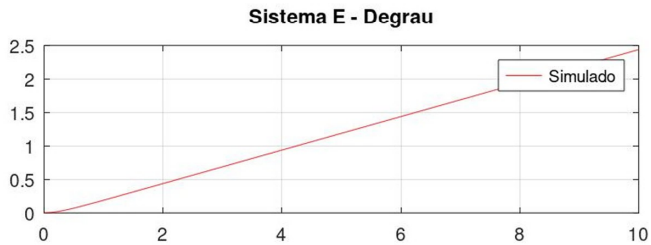
$u(t)$

$$Y(s) = \frac{1}{s} \cdot \frac{1}{s(s+4)}$$

$$\frac{1}{s^2(s+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+4}$$

$$Y(s) = -\frac{1}{16s} + \frac{1}{4s^2} + \frac{1}{16(s+4)}$$

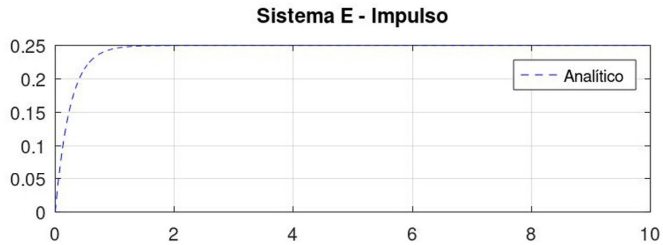


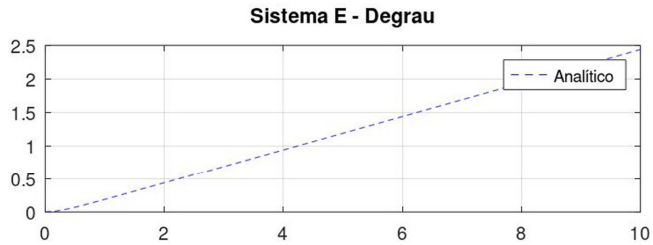
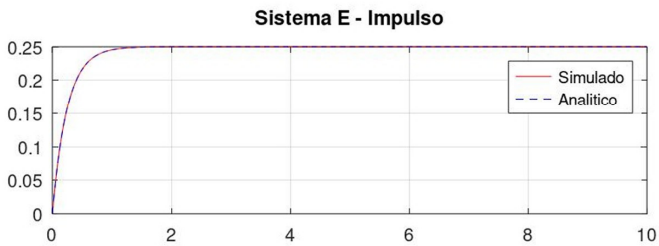
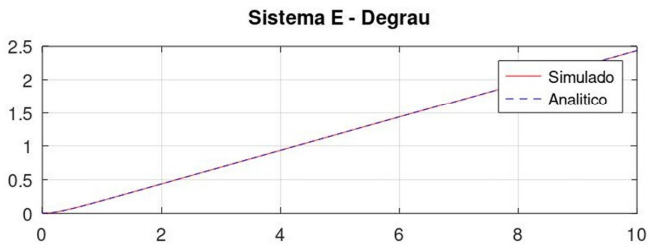


2. $y(t) = \frac{1}{4}(1 - e^{-4t}$

$$y(t) = \frac{t}{4} - \frac{1}{16} + \frac{1}{16}e^{-4t}$$

$$y(t) = \frac{t}{4} - \frac{1}{16}(1 - e^{-4t})$$

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